Reading GRPO training curves

Teach a Small AI to Think · unicodesnwrites.tech

Read curves like a doctor reads vital signs: **several together, as rolling averages, over tens of steps.** Single points are noisy, especially with only a few questions per step.

The vital signs

Vital sign	TRL name	Healthy	Worrying
Reward	`reward`, `rewards/<name>/mean`	rises, then flattens	flat at 0; or rising while held-out accuracy is not
Held-out accuracy	(you measure it)	rises with reward	flat or falling while reward rises
All-same groups	`frac_reward_zero_std`	well below 1; may creep up late as questions get solved	near 1 from the start: no signal
Reward spread	`reward_std`	stays above 0	shrinks to 0
Length	`completions/mean_length`	changes smoothly	slams into the limit, or crashes to a few tokens
Cut-off answers	`completions/clipped_ratio`	small	large or climbing
KL	`kl` (needs `beta > 0`)	small, smooth, slowly rising	sudden spikes; or exactly flat with no progress
Entropy	`entropy`	gentle decline as reward rises	sharp early drop with flat reward (collapse); steady rise (garbled text)
Clip ratio	`clip_ratio/region_mean`	about 0 with one update per batch; small otherwise	large or climbing

Four broken runs

Pattern	Giveaway	First fixes
Stuck at zero	reward ≈ 0, `frac_reward_zero_std` ≈ 1, nothing moves	read completions; test the grader by hand; raise `max_completion_length`; partial-credit format reward; easier data; raise `num_generations`
Reward hacking	reward up, held-out accuracy flat or down	stricter grader (one `<answer>`, one number); correctness worth more than format; cap shaping bonuses
Collapse	entropy crashes early, answers in a group identical, reward stalls	lower learning rate; `epsilon_high` (clip-higher); keep temperature ≈ 1.0; small `beta`; restart from a checkpoint
Leash too tight	slow progress, KL tiny, groups still disagree	lower `beta` or set it to 0
(bonus) **Rambling**	length climbs to the limit, accuracy flat	`loss_type="dapo"` or `"dr_grpo"`; `mask_truncated_completions=True`; cap any length bonus

Triage routine

1. **Read 10 real completions** (`log_completions=True` prints samples).
2. **Test each reward by hand** on right, wrong and messy answers you write.
3. **Check the fuel gauge**: `frac_reward_zero_std`.
4. **Compare reward with held-out accuracy.**
5. **Read entropy, length and KL together.**

6. **Change one knob at a time** and note every run.

See it on the Tiny Adder (real Python runs, seed 0, 200 steps)

Preset	Settings	Reward	Accuracy (true sum)
healthy	`G=8, lr=0.5, beta=0`	0.86	0.84
stuck	`G=2`	0.13	0.22
hacking	`buggy=True`	0.81	0.05
tight leash	`beta=0.05`	0.62	0.68

The in-browser version uses its own random numbers, so its values differ slightly. Collapse is not simulated: a perfectly graded lookup table can only become confident about the right answer.

Remember

- No disagreement in a group, no learning.
- Falling entropy is good only when reward is rising.
- The training reward cannot catch hacks of itself. Keep a held-out test.